# Stefan Vilceanu

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## Education

**RWTH Aachen University - M.Sc. Computer Engineering** Apr. 2023 – Mar. 2025
*Machine Learning, Embedded Systems, Computer Networks* *Aachen, DE*
Thesis: Uncertainty Estimates for Transformer-based Radar-, Camera-, and LiDAR-driven 3D Object Detectors

**RWTH Aachen University - B.Sc. Electrical Engineering** Oct. 2019 – Mar. 2023
*Electrical Circuits, Telecommunications, Embedded Systems* *Aachen, DE*

## Experience

**Porsche Engineering** Apr. 2025 – Present
*Software Engineer* *Stuttgart, DE & Timisoara, RO*
- Designed and implemented an AI Automation Agent for testing the UI of Porsche's Infotainment Systems
- Prototyped a real-time Speech AI Agent for testing the Digital Assistant of Porsche's Infotainment Systems
- Improved the existing testing framework by creating an AI Resolver to identify and repair broken testcases

**ETAS GmbH** Dec. 2024 – Apr. 2025
*Software Engineer Intern* *Stuttgart, DE*
- Analyzed real-world cyberattacks on vehicles and explored how existing SIEM systems could detect them
- Built a prototype transformer-based language model to generate accurate security queries from natural language

**IKA RWTH Aachen** Apr. 2024 – Nov. 2024
*Computer Vision Research Assistant* *Aachen, DE*
- Developed a Lane Analysis Algorithm using YoloPv2, Inverse Perspective Mapping and Bayesian Filtering
- Implemented ROS2 nodes to enable algorithm deployment on a real testing truck

**WZL RWTH Aachen** Mar. 2023 – Apr. 2024
*Robotics Research Assistant* *Aachen, DE*
- Developed and trained deep learning models to predict torque in robot joints
- Implemented MATLAB scripts to generate robot trajectories for robot experiments on the MABI 100

**IAEW RWTH Aachen** Jun. 2021 – Jul. 2022
*Smart Grids Research Assistant* *Aachen, DE*
- Developed a Python framework to visualize hybrid electrical grids using OpenStreetMap
- Designed the planning of hybrid AC/DC grids using the Genetic algorithm

## Projects

**Inference Pipeline for an Autonomous Car** | *C++, RPi5, HailoRT, UART, PID Control, Sensor Fusion*
- Real-time C++ perception-to-actuation pipeline on RPi 5 + Hailo-8L, using YOLOv8 and Depth Anything V2
- Implemented person-following autonomy using SORT tracking with PD/PI control from depth data
- Added hands-free voice control via custom ONNX wake-word models streamed into the pipeline

**Autonomous FPV Drone** | *Betaflight, ArduPilot, ELRS, Flight Automation, Flight Controller, Waypoint Navigation*
- Built an FPV drone from scratch, flashing and configuring ArduPilot, ELRS and the video system
- Tuned and improved flight performance through blackbox data logging and PID analysis
- Configured GPS-based autonomous waypoint navigation using ArduPilot, enabling preplanned missions

**Autonomous UAV Simulation – Ongoing** | *Jetson Orin, Gazebo, QGC, ROS2, PX4, Docker, TensorRT, VSLAM*
- Architecting an autonomous UAV stack split across a WSL workstation and a Jetson Orin Nano
- Building a six-phase perception and guidance pipeline including TensorRT-accelerated detection, Isaac ROS cuVSLAM visual-inertial odometry and min-jerk trajectory generation with L1 guidance

## Technical Skills

**Languages**: C++, Python, C, Bash/Shell, MATLAB
**Robotics & Autonomy**: ROS2, PX4, Betaflight, ArduPilot, Gazebo, VSLAM, Control Theory, Estimation, Filtering
**AI & Computer Vision**: PyTorch, OpenCV, Scikit-learn, HF Transformers, LangChain/Graph
**Edge AI & Deployment**: NVIDIA Jetson, TensorRT, ONNX, HailoRT, Docker, FastAPI
**Tools & Systems**: Linux, WSL, Simulink, Git, GitLab CI/CD, Jenkins, Nginx, Weights & Biases